None of these slingshots looked like the ones I described above. Instead of strongly deflecting the spacecraft's direction of motion, Venus, Earth, Jupiter, and Io deflected it only mildly. Why?

The deflectors' gravity was too weak to produce a strong deflection. For Venus, Earth, and Io, the deflection was inevitably small because their gravity is intrinsically weak. Jupiter has much stronger gravity, but a large deflection would have sent *Cassini* in the wrong direction; reaching Saturn required a small deflection.

Despite the small deflections, *Cassini* got substantial kicks from the flybys, big enough to compensate for inadequate fuel. In each flyby (except Io), *Cassini* traveled behind the deflecting planet but at an angle, so the planet's gravity optimally pulled *Cassini* forward, speeding it up. In *Interstellar*, the *Endurance* does a similar slingshot around Mars.

Cassini has been exploring Saturn and Saturn's moons for the past ten years, sending back amazing images and information—a treasure trove of beauty and science. For a glimpse, see http://www.nasa.gov/mission_pages/cassini/main/.

By contrast with these weak slingshots in the solar system, Gargantua's intense gravity can grab even objects moving at ultrahigh speeds and throw them around on strongly bent slingshots. Even a light ray. This produces gravitational lensing, the key to seeing Gargantua.

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19 The probability of finding IMBH's at the needed locations and times is small, but in the spirit of science fiction, since it is within the bounds of physical law, we can utilize them.